EASTERN VISAYAS STATE UNIVERSITY – ORMOC CAMPUS
COMPUTER STUDIES DEPARTMENT

TRIMINDER: DEVELOPMENT OF A GAMIFIED SCREEN TIME MANAGEMENT AND WELLNESS MONITORING APPLICATION FOR PROMOTING HEALTHY DIGITAL HABITS AMONG EVSU ORMOC STUDENTS

A Capstone Project Proposal
Presented to the Faculty of the Computer Studies Department
In Partial Fulfillment of the Requirements
for the Degree
Bachelor of Science in Information Technology

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APPROVAL SHEET

This Capstone Project Proposal entitled:

“TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits among EVSU Ormoc Students”

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ABSTRACT

This Capstone Proposal presents the development of TriMinder, a development of a gamified screen time management and wellness monitoring application for promoting healthy digital habits among Eastern Visayas State University - Ormoc Campus Students. The project addresses the shortcomings of existing digital wellness applications, which often rely on static and dismissible reminders that fail to encourage behavioral change. TriMinder aims to introduce a more interactive and persuasive approach by incorporating automated screen break enforcement and gamified incentives such as progress tracking and rewards. The system is designed for Android smartphones and is tailored to meet the needs of individual users. Utilizing the Agile development methodology, the application will be built iteratively with continuous user feedback, ensuring a user-centered design. Core features include screen usage monitoring and non-restrictive alerts. A trial deployment will assess user compliance, screen time reduction, and satisfaction. TriMinder is expected to enhance digital wellness through behaviorally informed engagement strategies. Its significance extends to users, offering a practical tool for fostering conscious screen use and reducing technology-related strain. Additionally, it contributes to the field of information technology by applying gamification and behavioral science principles in mobile app development, serving as a foundational study for future research in digital wellness solutions.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

As digital technology becomes increasingly integral to daily life, mobile devices have emerged as central components of students' academic, social, and personal activities. The Philippines ranks among the top three countries globally for highest screen time, with Filipinos spending an average of 8 hours and 52 minutes daily on digital devices—significantly exceeding the global average of 6 hours and 43 minutes (Philstar, 2024). This extensive digital engagement is particularly pronounced among college students, with research indicating that Filipino college students are among the heaviest users of social media in the world, with an average of nine hours per day spent on these platforms (Kemp, 2019).

Statistical data reveals concerning patterns among Filipino college-aged populations (18-24 years old). The Internet penetration rate in the college-aged (18–24 years old) population is 86% (Statista, 2022), indicating widespread digital connectivity. However, out of the 105.7 million Philippine population, 67 million (63%) have Internet access, with an average of 9 hours and 23 minutes Internet usage every day (Mateo, 2018). These statistics demonstrate a demographic skew toward higher digital consumption among younger populations, particularly college students.

Research has established that excessive screen time among college students correlates with adverse health outcomes. A comprehensive study on digital stress among Filipino college students documented significant impacts on sleep quality, academic performance, and physical well-being (Tandfonline, 2024). Similarly, international research confirms that prolonged screen exposure disrupts circadian rhythms, reduces attention spans, and contributes to digital eye strain (WHO, 2023). These findings substantiate concerns regarding students' inability to self-regulate their digital consumption patterns.

At Eastern Visayas State University (EVSU) Ormoc Campus, preliminary observations indicate that students across various departments demonstrate high dependency on smartphones for both academic and recreational purposes. Evidence of inadequate self-discipline in screen time management emerges from the documented correlation between excessive digital device usage and declining academic productivity, as supported by studies showing that excessive screen time is a significant factor in student mental health crises (THE Journal, 2024).

Current digital wellness interventions demonstrate significant limitations in addressing these challenges. Existing screen time monitoring applications, such as Digital Wellbeing and third-party alternatives, primarily function as passive tracking tools without incorporating motivational mechanisms or community engagement features. Research indicates that while gamification can improve time management and productivity, reliance solely on gamification is insufficient due to limited motivational impact (SpringerLink, 2024), highlighting the need for comprehensive approaches that integrate multiple engagement strategies.

This capstone project proposes the development of TriMinder, a gamified screen time management and wellness monitoring application specifically designed for Android devices. The system addresses identified gaps by incorporating evidence-based gamification principles, community-driven accountability mechanisms, and administrative oversight capabilities. The application will be supported by a web-based administrative platform, with the Student Affairs and Services Office (SASO) serving as super administrators to ensure institutional alignment and governance consistency.

1.2 Problem Statement

College students at EVSU Ormoc Campus exhibit excessive and unregulated screen time patterns that negatively impact academic performance, sleep quality, and overall wellness. Current digital wellness tools lack engagement mechanisms, gamification features, and community integration necessary for sustained behavior modification. No centralized platform exists to enable institutional monitoring of digital wellness trends or to facilitate recognition programs for students demonstrating healthy digital habits. This absence of structured, campus-specific digital wellness intervention represents a critical gap in student support services.

1.3 General Objectives

To develop a gamified screen time management and wellness monitoring application for promoting healthy digital habits among EVSU Ormoc College students.

1.4 Specific Objectives

- To design and develop an Android mobile application that enables students to monitor application usage, establish personal goals, and receive wellness-oriented notifications and recommendations.
- To implement gamification features—defined as the application of game-design elements such as points, badges, and leaderboards in non-game contexts (Deterding et al., 2011)—including experience points (XP), digital discipline badges, and performance-based reward systems.
- To construct a comprehensive ranking system accessible to students, SASO administrators, and department-based sub-administrators for recognizing students demonstrating exemplary digital discipline practices.

- To develop user engagement features including friend connection capabilities and peer interaction mechanisms to foster community-based wellness accountability.
- To create a web-based administrative platform for SASO that provides system oversight, user management capabilities, and institutional wellness analytics.
- To enable department-based organizations to access filtered ranking data and implement recognition initiatives promoting healthy digital habits within their respective academic units.
- To conduct comprehensive usability evaluation based on ISO/IEC 25010 software quality standards to ensure system effectiveness and user satisfaction.

1.5 Scope Limitations

Scope

This project encompasses the development of:

- An Android mobile application providing:
    - Screen usage monitoring and analysis
    - Gamified wellness notifications and break reminders
    - Achievement system with XP points and digital discipline badges
    - Peer connection and interaction capabilities
    - Personal goal setting and progress tracking features
- A web-based administrative platform enabling:
    - SASO super administrators to oversee campus-wide digital wellness metrics
    - Department-based sub-administrators to access filtered organizational data
    - Individual student profile management and recognition systems

    - Comprehensive reporting and analytics capabilities
- Integration features including:
    - Cloud-based data synchronization
    - Cross-platform data consistency
    - User authentication and security protocols

Limitations

The application development is constrained by several technical and methodological factors:

Platform Limitation: The system is exclusively developed for Android devices (API Level 24 and above) due to development resource constraints and the predominant Android usage among the target population (79.1% based on preliminary survey data). This platform-specific approach enables optimization of device-specific features such as usage access permissions and notification systems while maintaining development feasibility within academic timeline constraints.

Offline Functionality: Core features require internet connectivity for complete functionality, though basic tracking operates offline with synchronization upon connection restoration.

Administrative Access: Recognition and administrative features are limited to verified SASO personnel and authorized department representatives to maintain institutional governance standards.

Data Privacy: User data collection is restricted to usage patterns and wellness metrics, with personal content access explicitly prohibited to ensure privacy compliance.

Evaluation Scope: Usability testing will be conducted with a representative sample of students across departments due to time and resource limitations inherent in academic project timelines.

1.6 Significance of the Study

This research addresses critical digital wellness challenges facing contemporary higher education students. The primary beneficiaries, EVSU Ormoc College students, will gain access to evidence-based tools for developing sustainable digital habits through gamified engagement strategies and peer-supported accountability mechanisms.

Research demonstrates that college counterparts may prefer more complex games with progressive difficulty levels and social features for collaboration or competition (British Journal of Educational Technology, 2024), supporting the project's focus on sophisticated gamification elements. Additionally, studies indicate that gamification applied with an appropriate study method significantly improves time management in university students, and positively influences their academic results (Tandfonline, 2024).

The Student Affairs and Services Office (SASO) and department-based organizations will benefit through enhanced capacity to monitor student wellness trends, implement recognition programs, and develop data-driven intervention strategies. This administrative capability aligns with institutional mandates for comprehensive student support services.

Secondary beneficiaries include faculty members, guidance counselors, and institutional researchers who may utilize aggregated wellness data for evidence-based program development. The system's contribution to institutional digital transformation supports broader modernization initiatives while providing practical applications for information technology education.

The project's technological significance extends to the demonstration of gamification principles in educational technology applications, contributing to the growing body of research on digital wellness interventions in higher education contexts. This work supports the Commission on Higher Education's advocacy for digital wellness and academic productivity in Philippine higher education institutions.

1.7 Definition of Terms

Gamification – The application of game-design elements and principles (such as points, badges, leaderboards, and challenges) to non-game contexts to enhance user engagement and motivation (Deterding et al., 2011).

Screen Time – The cumulative duration of active interaction with digital device displays, typically measured in hours per day and categorized by application or platform usage.

Digital Discipline – The practice of conscious, regulated digital device usage that prioritizes wellness, productivity, and balanced lifestyle maintenance over compulsive or excessive consumption patterns.

Experience Points (XP) – A quantified measurement system used in gamification to track user progress and achievements, awarded for demonstrating positive digital wellness behaviors within the application.

SASO – Student Affairs and Services Office, the institutional administrative unit responsible for comprehensive student support services, serving as the super administrator for the TriMinder system.

Sub-administrator – Department-based personnel granted limited administrative privileges to access filtered student wellness data specific to their organizational unit.

ISO/IEC 25010 – An international standard framework for evaluating software system quality based on characteristics including functionality, usability, reliability, performance efficiency, and security.

Digital Wellness – A holistic state of balanced technology use that promotes physical health, mental well-being, and productive engagement with digital tools without negative impacts on sleep, social relationships, or academic performance.

CHAPTER 2: REVIEW OF RELATED LITERATURE

2.1 Introduction

This chapter presents a comprehensive thematic review of literature and empirical studies relevant to screen time behavior among college students, gamified mobile applications for wellness interventions, peer-supported digital discipline systems, and administrative oversight mechanisms. The reviewed themes include: (1) Screen time behavior and academic consequences; (2) Digital wellness challenges in higher education; (3) Gamification effectiveness in mobile wellness applications; (4) Social interaction and peer influence in behavior change; (5) Existing screen time management applications and their limitations. The chapter synthesizes key findings to identify research gaps that justify the proposed TriMinder system.

2.2 Thematic Review of Literature

2.2.1 Screen Time Behavior and Academic Consequences

Recent empirical research has established significant correlations between excessive screen time and academic performance decline among college students. A comprehensive study examining digital screen time (DST) usage among university students found that excessive DST has been shown to impair learning abilities, weaken critical thinking skills, and hinder academic performance (Frontiers in Psychiatry, 2025). This finding directly supports the claims made in Chapter 1 regarding the impact of unregulated screen time on academic productivity.

Quantitative research utilizing official GPA and ACT/SAT scores from university records demonstrates that time spent using a smartphone, especially on social networking, reduces GPA (ScienceDirect, 2021). This empirical evidence validates concerns about students' inability to self-regulate their digital consumption patterns, as referenced in the background study.

A systematic review and meta-analysis examining screen media use and academic performance found that television viewing and video game playing seem to be the activities most negatively associated with academic performance, particularly among adolescents (PMC, 2019). The research specifically highlights the need for further investigation into mobile phone use associations with academic outcomes, supporting the relevance of mobile-focused interventions.

Contemporary research has also documented that excessive screen time can negatively impact college students' academic performance, leading to decreased focus, reduced productivity, and difficulties in retaining information (Honor Society, 2024). These findings provide substantial empirical support for the problem statement's assertion that excessive screen time affects academic performance.

2.2.2 Digital Wellness Challenges in Higher Education

The mental health implications of excessive screen time among college students have been extensively documented. Research on college student populations reveals that screen time significantly predicted higher anxiety, depression, and stress (PubMed, 2023). This finding substantiates claims about the broader wellness impacts beyond academic performance.

Educational research indicates that a majority of educators said students' learning challenges rose along with their increased screen time and that student behavior worsened with more screen time (EdWeek, 2022). This institutional perspective validates the need for campus-wide intervention approaches, supporting the rationale for administrative oversight features in the proposed system.

Studies examining the developmental impacts of excessive screen exposure demonstrate that too much time spent in front of a screen and multitasking with other media has been related to worse executive functioning and academic performance (PMC, 2023). This research provides neurological basis for intervention strategies targeting screen time regulation.

2.2.3 Gamification Effectiveness in Mobile Wellness Applications

Empirical research on gamification effectiveness in mobile applications provides strong support for the proposed intervention approach. Studies utilizing partial least squares regression analysis demonstrate that gamification increases user engagement through satisfaction of the needs for competence, autonomy and relatedness (ScienceDirect, 2021). This finding aligns with Self-Determination Theory principles underlying the TriMinder design framework.

Recent research on gamification in mobile applications reveals that rewards are perceived as the most effective gamification principle in stimulating customer engagement (SAGE Journals, 2025). This empirical evidence supports the inclusion of XP points, badges, and achievement systems in the proposed application design.

Comprehensive literature reviews examining gamification in mobile applications indicate that rewards are the most commonly used design element of gamification in mobile apps, with gamification having potential impact on psychology and behavior and providing positive benefits (SpringerLink, 2023). This finding validates the gamification approach proposed for addressing digital wellness challenges.

Research investigating gamification's impact on educational contexts demonstrates that gamification of education can enhance levels of students' engagement similar to what games can do, to improve their particular skills and optimize their learning (Smart Learning Environments, 2020). This evidence supports the application of gamification principles to student wellness interventions.

2.2.4 Social Interaction and Peer Influence in Behavior Change

Research examining the effectiveness of social features in gamified applications reveals that achievement and social interaction-related gamification features were positively associated with all three forms of brand engagement (emotional, cognitive and social) (ResearchGate, 2021). This finding provides empirical justification for the friend-based ranking and peer interaction features proposed in the TriMinder system.

Studies investigating specific game design elements demonstrate that social relatedness was positively influenced by avatars, a meaningful story, and teammates (ScienceDirect, 2016). This research supports the inclusion of peer connection capabilities and profile interaction features designed to foster social accountability.

The effectiveness of gamification elements in satisfying psychological needs has been empirically validated, with research showing that gamification increases user engagement through satisfaction of the needs for competence, autonomy and relatedness, leading to greater intention to use, disseminate word-of-mouth about, and to positively rate the app (ResearchGate, 2024). This evidence supports the comprehensive gamification approach proposed for the TriMinder system.

2.2.5 Limitations of Existing Digital Wellness Tools

Current digital wellness applications demonstrate significant limitations in addressing the complex needs of college students. Research examining existing screen time management tools indicates that traditional applications primarily function as passive tracking systems without incorporating motivational mechanisms or community engagement features. These tools lack the evidence-based gamification elements and peer interaction capabilities that research demonstrates are essential for sustained behavior change.

The absence of institutional integration in existing applications represents a critical gap in supporting campus-wide wellness initiatives. Current tools do not provide administrative oversight capabilities or department-based filtering mechanisms that would enable Student Affairs and Services Offices to implement comprehensive digital wellness programs.

2.3 Comparative Analysis of Screen Time Management Apps

Table 2.3.1 Comparison of Existing Screen Time Management Applications

Application	Platform	Key Features	Strengths	Limitation
Google Digital Wellbeing	Android	App usage timer, wind-down mode, dashboard	Built-in with Android simple UI	No gamification, no social features
StayFree	Android	Usage tracker, overuse alert, app locking charts	Lightweight and simple	No community-based features
Forest	Android/ iOS	Stay focused by planting virtual trees, a rewards-based system	Unique visual gamification promotes focus	Limited scope for wellness tracking
RescueTime	Android/ iOS	Detailed activity reports, goal setting, and productivity scoring	Advanced analytics, focus-oriented	Lacks gamification and peer recognition
Space	Android/ iOS	Goal setting, screen time reduction coach, insights	Personalized insights	Subscription-based, no social sharing features
YourHour	Android	Time tracking, daily reports, addiction level scoring	Personalized addiction scoring	Too many ads, some features are premium only

Flipd	Android/ iOS	Full phone lockouts, wellness content, focus sessions	Strict mode encourages discipline	Invasive for some users, limited social interactivity
ScreenZen	Android/ iOS	Self-reflection prompts, mindfulness break, time limits	Combines mental health with screen tracking	Minimal gamification, limited data visualization
LockStock	Android	Rewards for locking the phone during class	Incentivizes offline time in campus contexts	Only works within defined geofences

The comparative analysis reveals that existing applications lack the comprehensive integration of gamification, peer interaction, and administrative oversight capabilities that research demonstrates are essential for effective behavior change interventions in educational contexts.

2.4 Theoretical and Conceptual Framework

Theoretical Foundation

Self-Determination Theory (SDT) serves as the primary theoretical foundation for the TriMinder system design. SDT posits that intrinsic motivation is enhanced when interventions address three basic psychological needs: autonomy (user control over goals and settings), competence (achievement through XP points and badges), and relatedness (peer connections and social recognition) (Deci & Ryan, 1985).

Research validation of SDT in gamification contexts demonstrates that gamification increases user engagement through satisfaction of the needs for competence, autonomy and relatedness (ScienceDirect, 2021), providing empirical support for the theoretical framework underlying the proposed system.

Conceptual Framework

This capstone integrates the SDLC (Software Development Life Cycle) with SDT to model system flow:

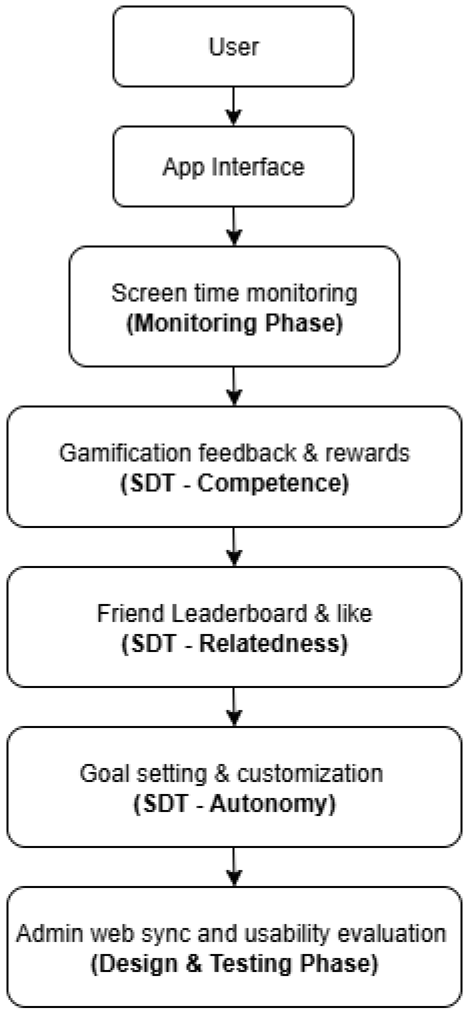

Figure 2.4.1 *Conceptual Framework of TriMinder Based on SDLC and Self-Determination Theory (SDT)*

2.5 Synthesis and Research Gap

The comprehensive literature review reveals significant empirical support for the claims made in Chapter 1 regarding the negative impacts of excessive screen time on college student academic performance and wellness. Research consistently demonstrates that unregulated digital device usage correlates with decreased academic achievement, increased mental health challenges, and impaired cognitive functioning.

Evidence from gamification research validates the proposed intervention approach, with multiple studies confirming the effectiveness of reward-based systems, peer interaction features, and achievement mechanisms in promoting sustained user engagement and behavior change. The integration of social features, particularly friend-based rankings and peer recognition systems, receives strong empirical support from research on social relatedness and community engagement. However, several critical research gaps emerge from the literature analysis:

1. **Limited Campus-Centric Solutions**: Existing research focuses primarily on general population interventions rather than institution-specific approaches that integrate with student affairs and academic support systems.
2. **Insufficient Administrative Integration**: Current studies do not address the need for institutional oversight capabilities that enable Student Affairs and Services Offices to monitor wellness trends and implement recognition programs.
3. **Platform-Specific Research Gaps**: Limited research exists on Android-specific digital wellness interventions tailored to the predominant platform usage patterns among Filipino college students.

4. **Cultural Context Limitations**: Existing research primarily focuses on Western populations, with insufficient investigation of gamification effectiveness in Filipino higher education contexts.

The TriMinder system addresses these identified gaps by providing a campus-centric, culturally appropriate, and institutionally integrated digital wellness intervention that combines evidence-based gamification principles with administrative oversight capabilities.

CHAPTER 3: METHODOLOGY

3.1 Research Design

This study employs a developmental research design using a mixed-methods approach that integrates both qualitative and quantitative data gathering. The mixed approach is suitable for addressing the practical needs of system development, usability testing, and user-centered design. It enables the collection of rich, contextual data on user behavior and preferences while allowing statistical validation of trends and usability outcomes. The study is primarily developmental, as it focuses on the iterative creation, testing, and refinement of the "TriMinder" mobile and web-based application for screen time management and wellness monitoring.

This design is aligned with the project's aim of building a gamified digital wellness platform specifically tailored to the needs of Eastern Visayas State University (EVSU) Ormoc students. Through user feedback loops and progressive system sprints under the Agile methodology, the design will ensure continuous improvement based on both empirical insights and user engagement data.

3.2 Data Collection Methods

Data for this study will be collected using structured survey questionnaires. These surveys will be distributed to the following respondents:

- EVSU Ormoc College students from any year level (general users)
- Student organizations per department (serving as sub-admins)
- Supreme Student Government (SSG) officers (super admin perspective)

A minimum of 100 student participants will be targeted using **purposive sampling**, focusing on those who are active smartphone users. Each department will have one officer to represent sub-admins. One or more representatives from the SSG (e.g., president or digital wellness officer) will be selected as key informants.

Among students surveyed, app usage patterns and preferred features like dark mode, wellness notifications, and non-intrusive ranking interactions were prioritized.

Survey distribution and consent:

- Student participants will receive an anonymous consent-based questionnaire either online or via printed forms. Participation is voluntary, and no identifiable personal data will be collected.
- Student organization officers and SSG respondents will be asked to sign a formal letter of informed consent acknowledging their participation and understanding of the project.

The data collected will cover:

- Digital usage behavior (daily screen time, types of apps used)
- Awareness and perceptions about digital wellness
- Preferences for gamification features (XP, badges, ranking, friend system)
- Perceived usefulness and interest in a wellness tracking system

Ethical Considerations

All participants will be informed about the research objectives and assured that their participation is entirely voluntary. They will have the right to withdraw at any time without consequence. Informed consent will be obtained from all participants, and in the case of minors, parental or guardian consent will be secured before their participation. The study will not collect any personally identifiable information, and all responses will be treated with strict confidentiality.

Data will be stored securely and used solely for academic purposes, according to established ethical research standards.

3.3 Data Analysis Techniques

Quantitative data from student surveys will be processed using descriptive statistics, including frequencies, percentages, means, and standard deviation to understand digital behavior and design preferences. These statistics will help prioritize features such as which rewards, alerts, or ranking interactions are most engaging.

The qualitative responses (especially from the comment section) were categorized thematically (e.g., concerns about intrusiveness, ranking pressure, cross-device tracking, battery optimization, etc.).

Qualitative data from student organization and SSG respondents will be thematically coded into categories such as:

- Recognition practices for digital wellness
- Desired admin features and ranking controls
- Feedback mechanisms and motivational strategies

The combination of quantitative and qualitative analysis will guide system design decisions and provide a foundation for usability testing, validation of UI/UX elements, and enhancement of gamified engagement mechanisms.

3.4 System/Software Design

The **Agile Software Development Life Cycle (SDLC)** model will be used in this project. Agile is selected for its iterative nature, ability to accommodate ongoing feedback, and suitability for user-centered system design. It is particularly beneficial in academic projects that involve multiple stakeholders and require progressive refinement.

The Agile phases to be followed include:

- **Planning:** Define features based on research objectives and survey feedback.
- **Analysis:** Analyze survey data to determine user preferences and system requirements.
- **Design:** Create wireframes and interface mockups using user requirements.
- **Implementation:** Code mobile and web-based modules through sprint cycles.
- **Testing:** Perform unit and integration testing, and conduct user-based usability evaluations.
- **Deployment:** Prepare the application for pilot release within the EVSU Ormoc campus.
- **Maintenance:** Address feedback, bugs, or enhancements identified during testing.

3.5 Tools and Technologies

The development of TriMinder will utilize a combination of modern frameworks, backend services, and developer tools to ensure a responsive user experience, efficient code management, and scalable architecture.

- **Development Platform and Backend Services:**
    - **Supabase** will be used as the Backend-as-a-Service (BaaS) platform. It provides real-time database features, authentication, and RESTful APIs, making it suitable for managing user data, authentication, screen activity logs, and gamified rewards. Supabase's open-source PostgreSQL infrastructure ensures reliability and scalability throughout the application's lifecycle (Supabase, n.d.).

  - **SQLite** will also be integrated for local storage purposes. As a lightweight, embedded SQL database engine, SQLite is ideal for storing data directly on the user's device, providing offline access to essential data such as cached activity logs and user preferences. Its serverless design and zero-configuration setup make it efficient for mobile and web applications (SQLite, n.d.).
- **Programming Language and Framework:**
  - The system will be developed using **Flutter**, a cross-platform open-source UI toolkit developed by Google. Flutter enables the creation of natively compiled applications from a single codebase, ideal for building responsive, interactive, and gamified mobile applications. It supports high-performance rendering and a widget-based architecture well-suited for both children and adult users (Google Developers, n.d.).
- **Code Editor:**
  - **Visual Studio Code (VS Code)** will serve as the primary code editor. VS Code offers advanced extension support for Flutter and Dart, built-in debugging tools, Git integration, and code intelligence features, making it an efficient tool for team-based development and testing (Microsoft, n.d.).
- **Design and Modeling Tools:**
  - **Draw.io** (also known as diagrams.net) will be used for creating system architecture diagrams, wireframes, data flow diagrams, and Entity-Relationship Diagrams (ERDs). These visual tools aid in both planning and documenting the software development process (Diagrams.net, n.d.).

   - **Lucidchart** will be used as a visual diagramming tool to aid in the design and planning of system architecture, database schemas, and process flows. Its collaborative interface allows the development team to visualize relationships and workflows clearly, improving communication and accuracy in system design (Lucidchart, n.d.).
   - **Figma** will serve as the primary user interface (UI) and user experience (UX) design tool. It allows the team to design high-fidelity prototypes, wireframes, and mockups in a collaborative environment. Figma's cloud-based platform enables real-time design feedback and efficient iteration across multiple team members (Figma, n.d.).
- **Project Management and Collaboration:**
   - **GitHub** will serve as the project's version control and collaboration platform. Through GitHub, developers will manage code repositories, track issues, conduct code reviews, and enable continuous integration workflows, ensuring codebase integrity and project transparency across the team (GitHub Docs, n.d.).

CHAPTER 4: NEEDS ASSESSMENT AND REQUIREMENT ANALYSIS

4.1 Needs Assessment

Current Situation Analysis

Currently, students at Eastern Visayas State University (EVSU), Ormoc Campus, have no unified system to track and manage their screen time. While some rely on built-in tools like Digital Wellbeing, these apps lack motivation features, social engagement, and gamified elements that are proven to enhance user retention and behavior change.

Students commonly attempt manual self-regulation but is often ineffective due to the absence of peer accountability, feedback mechanisms, or structured interventions. Additionally, no administrative tools exist for the Supreme Student Government (SSG) or departmental student organizations to monitor digital behavior trends or promote wellness initiatives.

Several inefficiencies were identified:

- There is no incentive system for students to practice digital discipline.
- Lack of peer-based engagement in digital wellness efforts.
- Fragmented tracking tools that do not include department-based visibility.
- No cross-device tracking, with many students reporting higher screen time on laptops than phones.

Stakeholder Input

A. Quantitative Results from Student Survey

Out of over 200 student respondents from EVSU Ormoc Campus, the majority reported heavy daily screen usage, averaging between 5 and 8 hours per day, with many also admitting to poor focus and time management. Approximately 85% used mobile phones as their primary device, while a smaller percentage used laptops or tablets. Most popular apps used include Facebook, TikTok, YouTube, and Messenger, indicating intense exposure to time-consuming digital platforms.

Regarding digital wellness solutions, students overwhelmingly favored features like break notifications (91%), XP-based rewards (84%), and daily screen time goals (87%). Interestingly, a substantial number also supported public elements such as leaderboards (73%) and friends' visibility (70%), validating the inclusion of gamified and social accountability features in TriMinder.

Q1. Average Smartphone Usage per Day

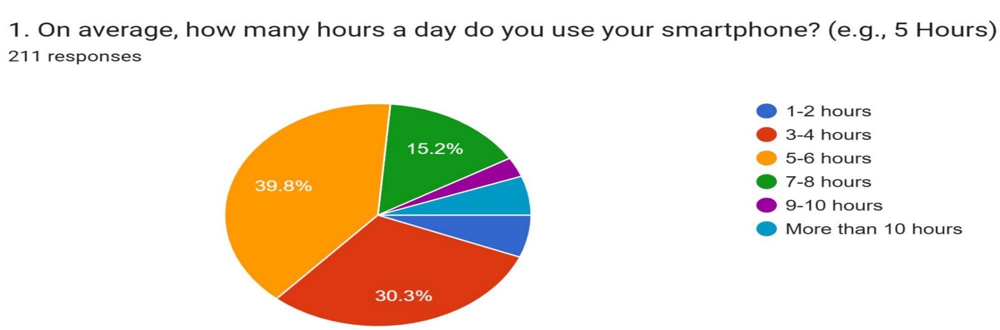

Figure 4.1.1 *Average Smartphone Usage per Day*

The data presented in Figure 4.1 highlights that the majority of student respondents—particularly those from EVSU Ormoc—reported using their smartphones between 5 and 6 hours daily. This prolonged screen time suggests a high level of dependency on digital devices, particularly for social, academic, and entertainment purposes. Such a pattern of usage supports the need for an intervention tool like TriMinder, which promotes screen time awareness and healthy digital habits through gamified features.

Q2. Primary Devices Used by Students

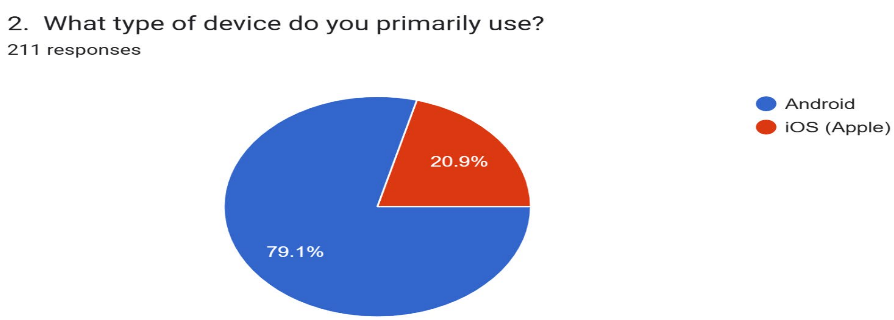

Figure 4.1.2 *Primary Devices Used by Students*

Figure 4.2 illustrates that 79.1% of respondents primarily use Android phones, followed by 20.0% who use iOS devices. This confirms that an Android-first approach is most appropriate for the proposed TriMinder application. Designing the system primarily for Android smartphones will ensure optimal accessibility and relevance to the majority of the target users.

Q3. Most Frequently Used Apps

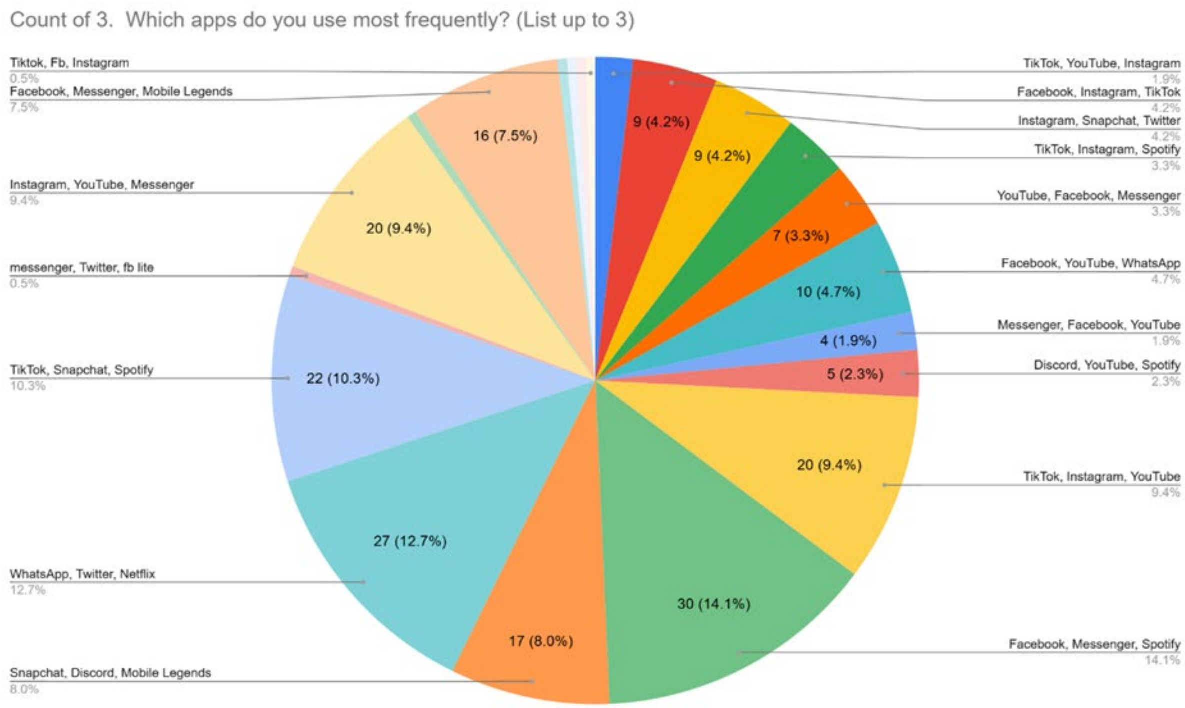

Figure 4.1.3 *Most Frequently Used Apps*

As shown in Figure 4.3, the top apps used by respondents include Facebook, TikTok, Messenger, YouTube, Instagram, and Spotify. These platforms are primarily for social interaction and entertainment, which may contribute to excessive screen time. TriMinder can address this trend by encouraging mindful usage through usage analytics and digital wellness rewards.

Q.4 Current Screen Time Tracking

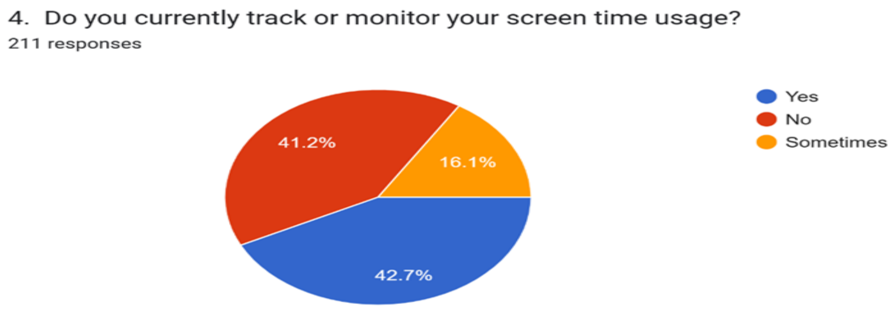

Figure 4.1.4 *Current Screen Time Tracking*

The results show that a large portion of students do not actively monitor their screen time, indicating low awareness of their digital habits. This gap suggests a strong opportunity for TriMinder to fill this role by providing user-friendly tracking features that raise awareness and encourage self-regulation.

Q5. Awareness of Screen Time Effects

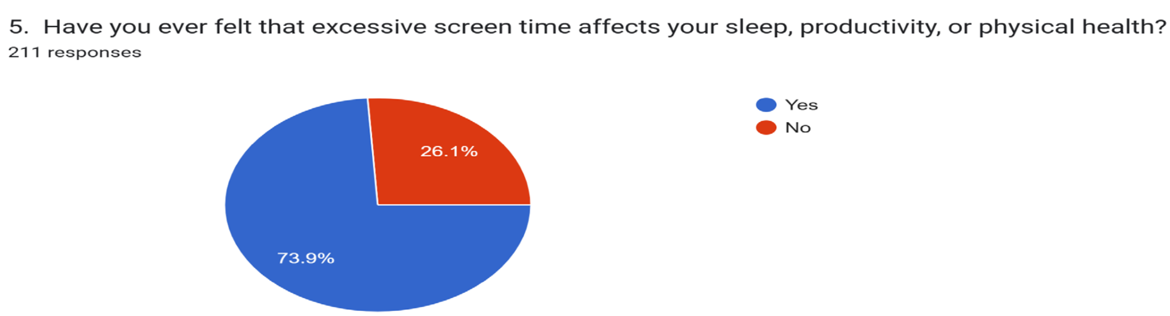

Figure 4.1.5 *Awareness of Screen Time Effects*

Figure 4.5 reveals that 73.9% of students acknowledge that excessive screen time negatively impacts their sleep, focus, or physical health. This data underscores the importance of wellness education and reinforces the relevance of TriMinder's health-focused features, such as break notifications and sleep-time reminders.

Q6. Willingness to Use a Wellness App

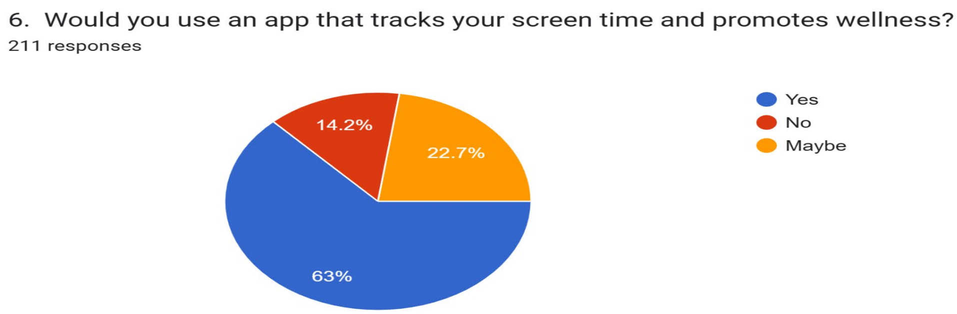

Figure 4.1.6 *Willingness to Use a Wellness App*

A majority of students expressed a positive response toward using an app that tracks screen time and promotes wellness. This validates TriMinder's overall concept and indicates strong user receptiveness to digital tools that support healthier technology use and lifestyle balance.

Q7. Importance of Break Reminders

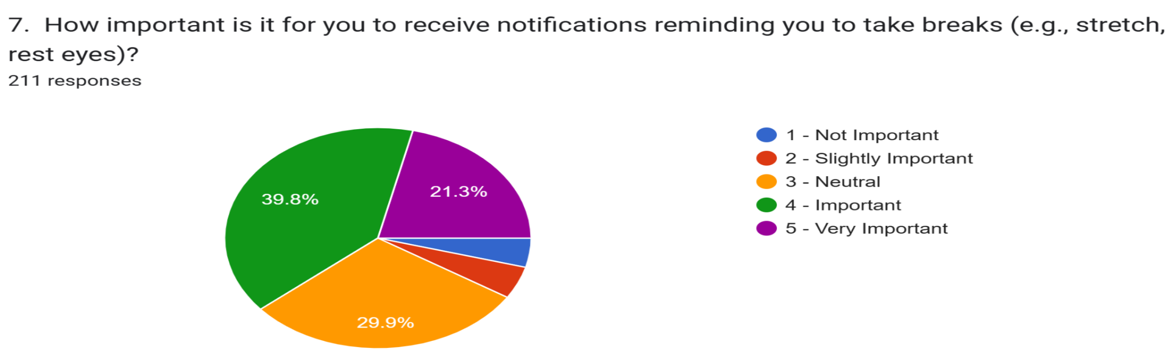

Figure 4.1.7 *Importance of Break Reminders*

According to the data, 91% of respondents find break notifications necessary, especially for activities like stretching or resting their eyes. This feature can serve as a key wellness component of TriMinder, helping prevent burnout and screen fatigue through timely and actionable prompts.

Q8. Motivation Through XP Points and Badges

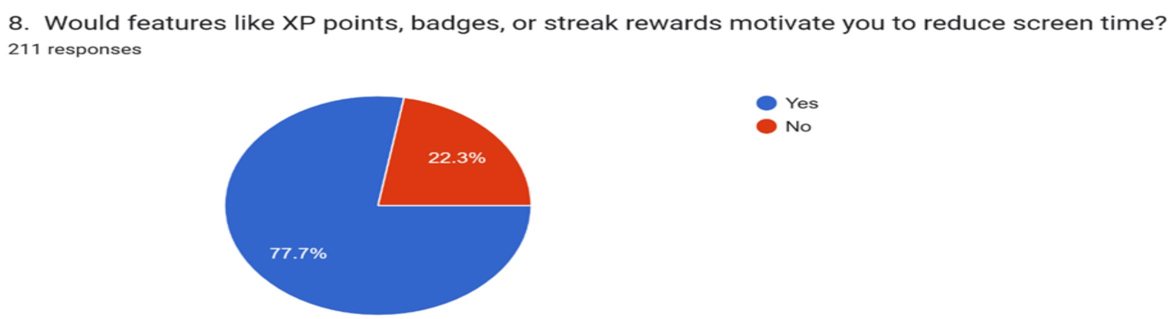

Figure 4.1.8 *Motivation Through XP Points and Badges*

The results show that 77.7% of students are motivated by gamification features such as XP points, badges, or streak rewards. This confirms that a gamified approach can be effective in promoting digital discipline, making wellness engagement more fun and sustainable within the TriMinder app.

Q9. Preferences for Personalization and Goal Setting

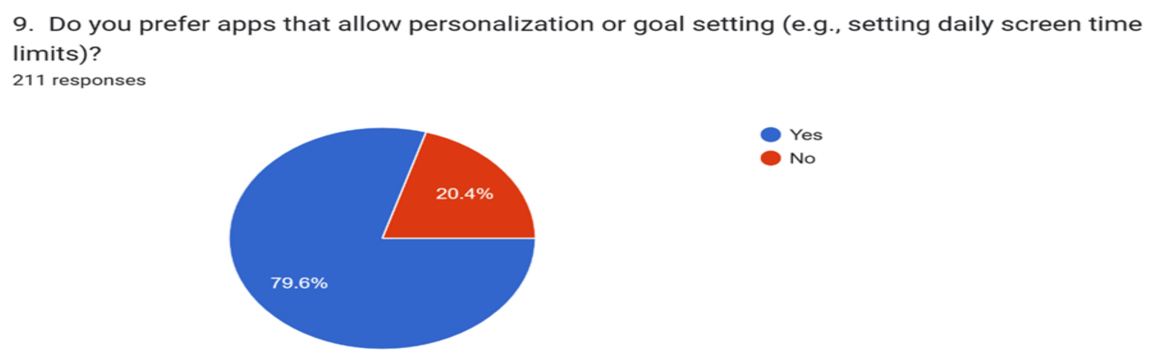

Figure 4.1.9 *Preferences for Personalization and Goal Setting*

A high 79.6% of respondents prefer apps that allow them to set personal goals or screen time limits. This indicates that customization is essential for user satisfaction, and TriMinder should include features that would enable students to tailor their experience to fit their unique habits and needs.

Q10. Motivation via Public Ranking

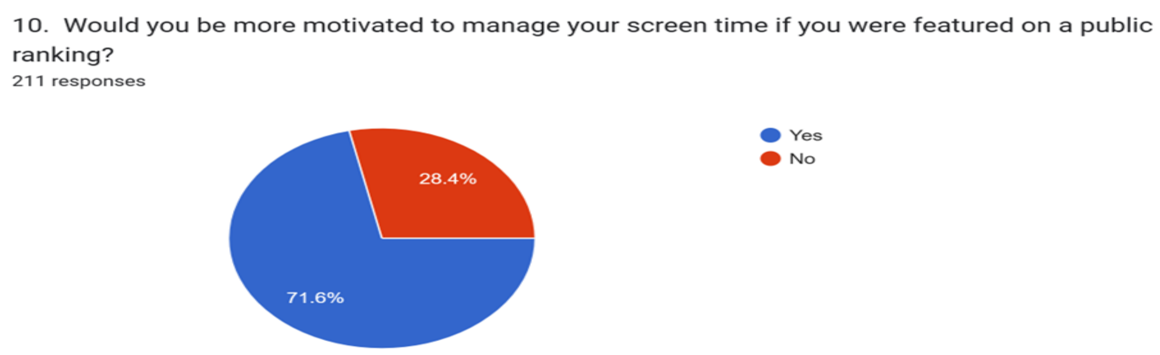

Figure 4.1.10 *Motivation via Public Ranking*

Figure 4.10 shows that 71.6% of students believe that being featured on a public ranking would motivate them to manage screen time better. This social visibility can drive friendly competition and accountability, making it a powerful feature for behavior change in the TriMinder app.

Q11. Viewing Friends' Progress

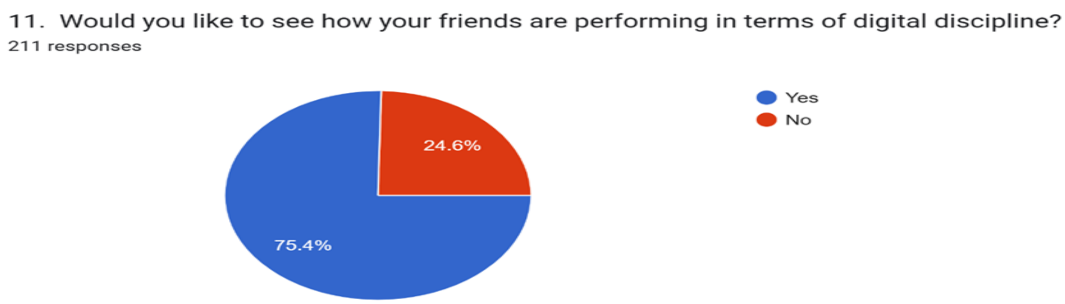

Figure 4.1.11 *Viewing Friends' Progress*

75.4% of students are interested in seeing their friends' progress in digital discipline, which supports the inclusion of community or peer interaction features in TriMinder. This can foster a sense of collective wellness and mutual encouragement among users.

Q12. Profile Like/Unlike Feature

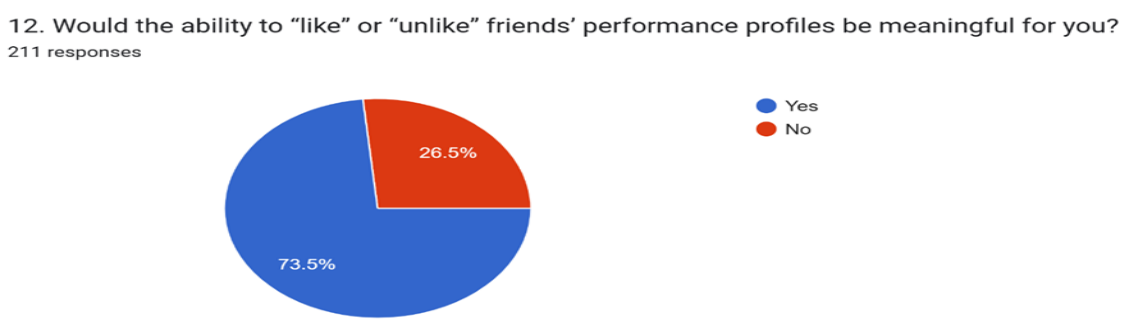

Figure 4.1.12 *Profile Like/Unlike Feature*

With 73.5% showing interest in the ability to 'like' or 'unlike' their friends' wellness profiles, this social feature could enhance user interaction and motivation. It introduces a simple way for students to acknowledge each other's progress, adding an extra layer of engagement in the app.

Q13. Trust in Student Leader for Recognition

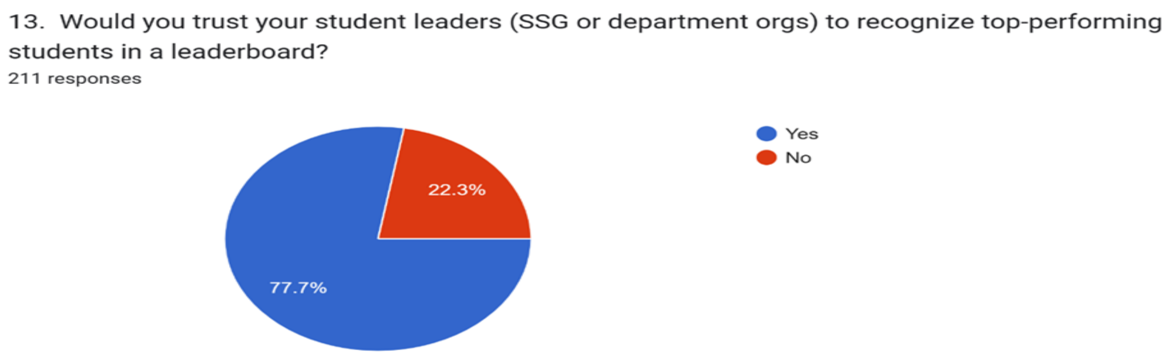

Figure 4.1.13 *Trust in Student Leader for Recognition*

A majority of students expressed trust in student leaders or organizations like the SSG to recognize top-performing users on a leaderboard. This shows a favorable attitude toward integrating wellness initiatives into student governance, adding credibility and incentive to TriMinder's leaderboard system.

Q14. Likelihood of Recommending the App

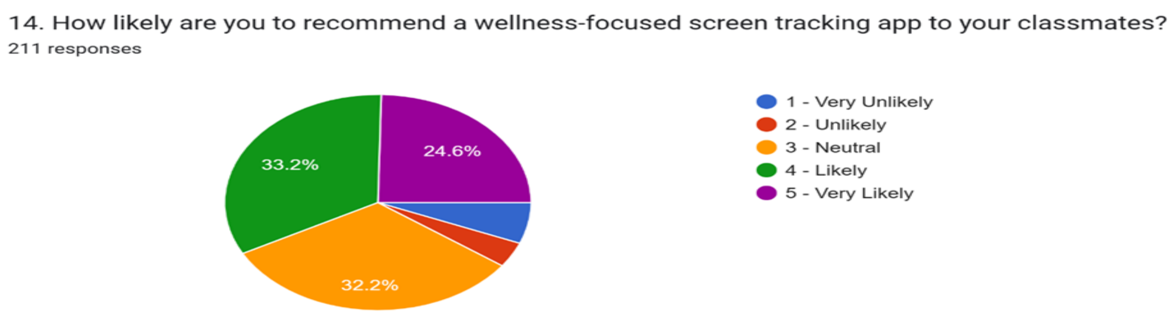

Figure 4.1.14 *Likelihood of Recommending the App*

Survey data shows that students are highly likely to recommend a wellness-focused screen tracking app like TriMinder to classmates, especially if it proves effective in promoting healthy habits. This reflects strong potential for organic app growth through peer endorsement and user satisfaction.

B. Thematic Analysis of Open-Ended Responses

Theme 1: Digital Overuse and Academic Strain

Many students described being unable to focus during study hours due to habitual app checking. The lack of control over their screen time leads to stress, cramming, and missed academic deadlines. Students frequently expressed frustration over procrastinating due to mindless scrolling, particularly at night or between classes.

Theme 2: Desire for Gamification

Students were notably receptive to gamified systems. They mentioned that achieving milestones, collecting XP, and receiving badges can positively reinforce better habits. Some suggested comparison features like "streaks" and "top 10 students per department" as added motivation.

Theme 3: Peer Influence and Social Comparison

The ability to see friends' progress and get "likes" for good behavior was perceived as a healthy form of motivation. Several responses highlighted that such visibility builds accountability and that a public leaderboard can encourage friendly competition.

Theme 4: Customization and Usability

Students emphasized the need for features like class-synced break reminders, battery-saving UI, and a low-storage footprint. Others expressed concern about privacy and proposed anonymized rankings to reduce pressure while still fostering motivation.

C. Stakeholder Interviews (SSG and Department Organization)

In discussions with SSG officers and department student organization leaders, the need for administrative features became very clear. The SSG emphasized the value of a system that enables non-intrusive monitoring, supports report generation, and celebrates digitally disciplined students through recognition programs or digital awards.

Student organization officers emphasized the desire to view filtered rankings by department, which could be used during student assemblies or intramurals to recognize balanced digital habits. They also raised concerns over data privacy and preferred view-only dashboards with anonymized data when needed.

Stakeholders widely supported a gamified app that would reward students, enable inter-department competition, and provide admin dashboards for easy tracking.

Problem-Solution Fit

There is an apparent disconnect between current practices and desired digital behavior outcomes. While students are aware of their excessive screen usage, they lack the tools, community support, and motivation to change.

The proposed TriMinder mobile app and admin web platform directly addresses these needs through:

- Gamified tracking of screen time behavior.
- XP points and badge rewards for digital discipline.
- Daily rankings and peer interaction via profile likes.
- Admin dashboards for SSG and sub-admins to view reports and promote wellness.
- Offline data caching and cloud sync for hybrid access.

These solutions align with international standards for usability and performance, and integrate user feedback collected through mixed-method research.

4.2 Requirement Analysis

Functional Requirements

- FR1: The system shall monitor and record daily screen time for Android users.
- FR2: The system shall award XP points and badges based on healthy digital habits.
- FR3: The system shall allow students to add friends and view "Friends Ranking."
- FR4: The system shall enable users to like or unlike peer profiles once per day.
- FR5: The system shall provide daily department-based and general student rankings.
- FR6: The system shall generate break notifications (e.g., stretch, rest eyes).

- FR7: The system shall allow SSG super admins to manage department sub-admins
- FR8: The admin platform shall allow sub-admins to view department-only reports.
- FR9: The system shall display individual user profiles with wellness stats and badges.
- FR10: The system shall allow syncing with cloud storage and offline SQLite caching.
- FR-11: The system shall export user screen time reports (e.g., before graduation).

Non-Functional Requirements

- NFR1: The system shall respond to user actions within 2 seconds.
- NFR2: The system shall maintain usability for students with basic mobile experience.
- NFR3: The system shall ensure secure login and data encryption.
- NFR4: The system shall be compatible with Android 8 and above.
- NFR5: The system shall consume less than 100MB of storage on mobile devices.
- NFR6: The system shall support dark mode and energy-efficient UI.
- NFR7: The system shall work offline and sync data when an internet connection is available.
- NFR8: The system shall be maintainable through modular code using Flutter and Supabase.

User Requirements

- Student Users: Can track screen usage, earn XP, add friends, like profiles, and view rankings.
- Sub-admin (Dept Org Officer): Can view department-specific rankings, encourage users via likes.
- Super Admin (SSG Officer): Can manage sub-admin access, view all student rankings, and recognize top users.

Table 4.2.1 System Requirements Table

Requirement ID	Type	Description	Priority
FR1	Functional	The system shall monitor and record daily screen time for Android users.	High
FR2	Functional	The system shall award XP points and badges based on healthy digital habits.	High
FR3	Functional	The system shall allow students to add friends and view "Friends Ranking.	Medium
FR4	Functional	The system shall enable users to like or unlike peer profiles once per day.	High
FR5	Functional	The system shall provide daily department-based and general student rankings.	High
FR6	Functional	The system shall generate break notifications (e.g., stretch, rest eyes).	Medium
FR7	Functional	The system shall allow SSG super admins to manage department sub-admins.	High
FR8	Functional	The admin platform shall allow sub-admins to view department-only reports.	Medium
FR9	Functional	The system shall display individual user profiles with wellness stats and badges.	Medium
FR10	Functional	The system shall allow syncing with cloud storage and offline SQLite caching.	High
FR11	Functional	The system shall export user screen time reports (e.g., before graduation).	Low
NFR1	Non-Functional	The system shall respond to user actions within 2 seconds.	High

NFR2	Non-Functional	The system shall maintain usability for students with basic mobile experience.	High
NFR3	Non-Functional	The system shall ensure secure login and data encryption.	Critical
NFR4	Non-Functional	The system shall be compatible with Android 8 and above.	High
NFR5	Non-Functional	The system shall consume less than 100MB of storage on mobile devices.	Medium
NFR6	Non-Functional	The system shall support dark mode and energy-efficient UI.	Medium
NFR7	Non-Functional	The system shall work offline and sync data when online.	High
NFR8	Non-Functional	The system shall be maintainable through modular code using Flutter and Supabase.	Medium
UR1	User	Students can track usage, earn XP, add friends, like profiles, and view rankings.	High
UR2	User	Sub-admins can view department-specific rankings and send likes.	Medium
UR3	User	Super admins can manage sub-admin access and recognize top users.	High

CHAPTER 5: PROJECT PLAN

5.1 Project Schedule

The development of TriMinder follows the **Agile Software Development Life Cycle (SDLC)** methodology, which allows for iterative development with continuous user feedback and progressive system refinement. The project is structured across **18 weeks** to align with the academic calendar and ensure thorough development, testing, and documentation phases.

Project Phases and Activities

The TriMinder project is divided into seven major phases, each containing specific activities and deliverables:

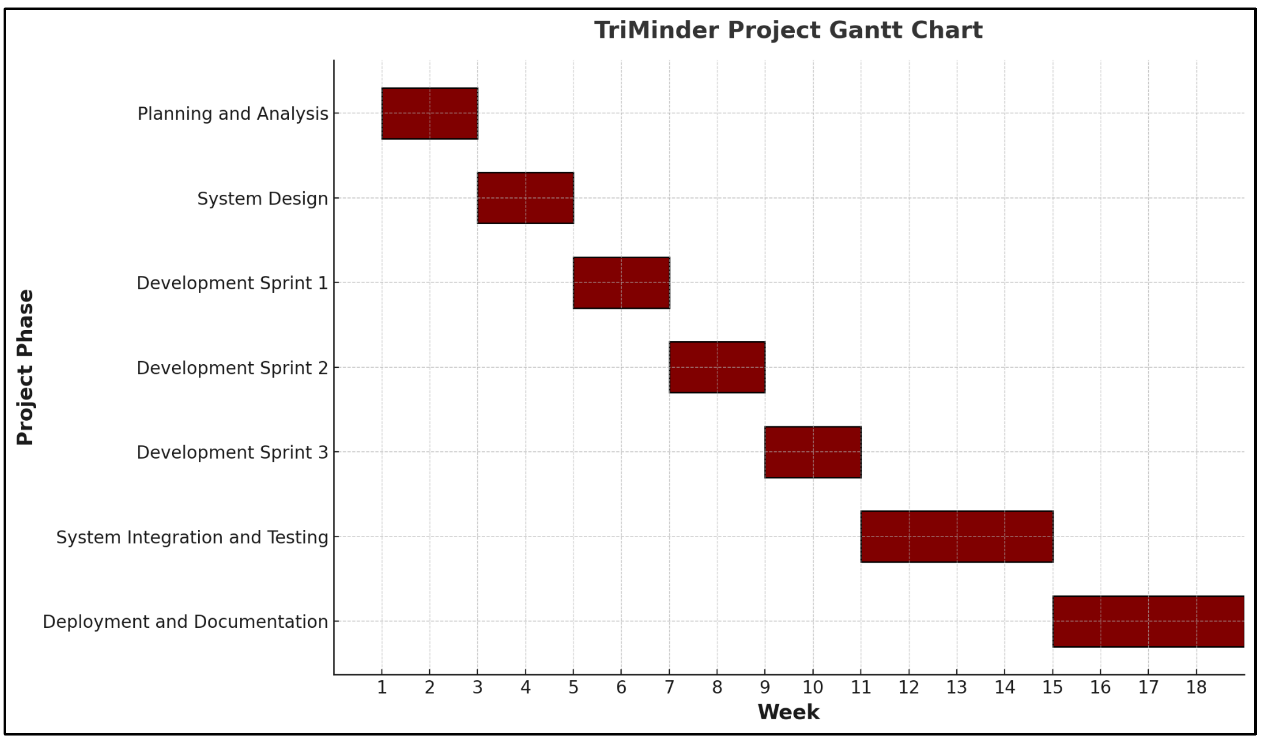

***Figure 5.1.1** Proposed Project Timeline for TriMinder (Week 1 to Week 18)*

Phase 1: Planning and Analysis (Weeks 1-2)

- Conduct comprehensive stakeholder surveys and interviews
- Perform literature synthesis and review existing solutions
- Finalize functional and non-functional requirements
- Establish project scope and limitations
- Complete needs assessment and requirement analysis

Phase 2: System Design (Weeks 3-4)

- Create system architecture and database design
- Develop Entity-Relationship Diagrams (ERD) and Data Flow Diagrams (DFD)
- Design wireframes and user interface mockups using Figma
- Establish UI/UX style guide and design standards
- Build low-fidelity prototypes for user validation

Phase 3: Development Sprint 1 (Weeks 5-6)

- Implement core screen time monitoring functionality
- Develop a user authentication system using Supabase
- Create a local database schema using SQLite
- Build fundamental tracking modules
- Conduct unit testing for core components

Phase 4: Development Sprint 2 (Weeks 7-8)

- Implement gamification engine (XP points, badges, streaks)
- Develop reward and achievement systems
- Create user dashboard and progress tracking interfaces
- Integrate a notification system for wellness breaks

- Conduct a sprint demo with pilot users

Phase 5: Development Sprint 3 (Weeks 9-10)

- Develop social features (friend system, ranking, likes)
- Implement an admin web platform for SSG and sub-admins
- Create department-based filtering and reporting
- Add real-time synchronization between mobile and web platforms
- Integrate all system modules and conduct integration testing

Phase 6: System Integration and Testing (Weeks 11-14)

- Perform comprehensive system integration
- Conduct functional, security, and performance testing
- Execute usability testing based on ISO/IEC 25010 standards
- Gather user feedback and iterate on UI/UX improvements
- Address identified defects and optimization issues

Phase 7: Deployment and Documentation (Weeks 15-18)

- Prepare application for Android deployment
- Complete comprehensive project documentation
- Conduct final system evaluation and metrics compilation
- Prepare for oral defense and panel presentation
- Execute project turnover and maintenance planning

Duration and Dependencies

- Critical dependencies have been identified to ensure smooth project progression:
- Sprint 2 cannot commence until Sprint 1 delivers stable API foundations and user authentication systems

- Integration Testing (Weeks 11-12) requires completion of all three development sprints
- Usability Testing (Weeks 13-14) depends on a stable beta build from the integration phase
- Deployment Preparation (Week 15) requires finalized UI/UX elements and branding assets from the Design phase
- Final Documentation cannot be completed until all testing phases provide comprehensive evaluation results

5.2 Milestones and Deliverables

The TriMinder project defines **nine major milestones** that serve as critical checkpoints for project progress evaluation and quality assurance. Each milestone produces specific, measurable deliverables that contribute to the overall project success.

Table 5.2.1 Milestone Table

Milestone No.	Description	Key Deliverable	Target Week	Success Criteria
M1	Requirements Analysis Completion	Requirements Specification Document	Week 2	Approved functional and non-functional requirements with stakeholder sign-off
M2	System Design Approval	ERD, DFD, and Wireframes Package	Week 4	Complete system architecture with adviser approval
M3	Core Functionality Demo	Functional Prototype (Tracking + Authentication)	Week 6	Working screen time monitoring with user login capabilities
M4	Gamification System Demo	Integrated Prototype with Rewards System	Week 8	Functional XP points, badges, and achievement tracking

M5	Social Features Integration	Complete Mobile App with Admin Web Platform	Week 10	Working friend system, ranking, and admin dashboard
M6	System Integration Completion	Integrated Beta Release	Week 12	Fully integrated system with zero Priority 1 bugs
M7	Usability Evaluation Sign-off	ISO/IEC 25010 Usability Report	Week 14	System Usability Scale (SUS) score ≥ 80 with positive user feedback
M8	Deployment Readiness	Release Candidate APK and Web Platform	Week 15	Production-ready applications with security validation
M9	Project Completion & Oral Defense	Final Manuscript, User Manual, Signed Turnover	Week 18	Approved final manuscript with successful oral defense

Risk Mitigation and Contingency Planning

Each milestone incorporates buffer time and alternative approaches to address potential delays:

- **Technical Risks**: Backup development approaches identified for complex integrations
- **Resource Risks**: Cross-training among team members to ensure continuity
- **Schedule Risks**: Priority-based feature implementation allowing scope adjustment if necessary
- **Quality Risks**: Continuous testing and integration throughout development phases

CHAPTER 6: TECHNICAL SPECIFICATION

6.1 Hardware Requirements

The TriMinder application requires specific hardware for development and user operation. The following specifications ensure optimal performance across all system components.

Table 6.1.1 Development Hardware

Component	**Minimum Specification**	**Recommended Specification**	**Purpose**
Developer Computer	Intel Core i5, 8GB RAM, 256GB SSD	Intel Core i7, 16GB RAM, 512GB SSD	Development and testing
Android Phone	Android 7.0+, 3GB RAM, 32GB storage	Android 10.0+, 6GB RAM, 64GB storage	App testing
Internet Connection	5 Mbps broadband	25 Mbps broadband	Cloud services access

6.2 Software Requirements

TriMinder's development leverages open-source tools and industry-standard frameworks to ensure scalability, maintainability, and cross-platform compatibility. The following tools and technologies are required throughout the application lifecycle.

Table 6.2.1 Development Software

Software/Tool	**Version**	**Function**
Flutter SDK	V3.24.0 or later	Mobile app development framework
Android Studio	Latest Version	Development Environment

Visual Studio Code	Latest Version	Code Editor (Alternative)
Git	Latest Version	Version Control
Android SDK	API Level 24-34	Android Development Tools

Table 6.2.2 Backend Services

Service	**Type**	**Function**
Supabase	Cloud-hosted	Database and authentication
PostgreSQL	Database (via Supabase)	Cloud data storage
SQLite	Local Database	Offline data storage and caching
Supabase Auth	Authentication service	User login and security
Supabase Storage	File storage	User data and app assets

Table 6.2.3 Design Tools

Tool	**Type**	**Function**
Figma	Web-based	UI/UX design
Draw.io	Web-based	System diagrams
Lucidchart	Web-based	System architecture and flowchart diagrams
GitHub	Cloud service	Code repository

Table 6.2.4 Target Platform

Platform Component	Requirement	Purpose
Android OS	Version 7.0 (API 24) minimum	App runtime environment
SQLite	Version 3.31+	Local database operations
Internet Browser	Chrome 70+ or equivalent	In-app web content

6.3 Network Requirements

TriMinder requires internet connectivity for synchronization, authentication, and cloud features. The application also supports offline functionality for core features through SQLite local storage.

Table 6.3.1 Internet Connectivity

Requirement	Specification	Purpose
Minimum Speed	1 Mbps download, 512 Kbps upload	Basic app functionality
Recommended Speed	5 Mbps download, 2 Mbps upload	Optimal performance
Connection Type	Wi-Fi or mobile data (4G/5G)	Flexible connectivity
Data Usage	10-50 MB per month	Estimated monthly consumption

Table 6.3.2 Network Features

Feature	Specification	Benefit
Offline Mode	Core features work without internet	Uninterrupted usage
Auto-Sync	Automatic data synchronization when online	Data consistency between local and cloud
Security	HTTPS/TLS encryption for all data	Secure communication
Cloud Backup	Automatic backup to Supabase Pro	Data protection and recovery

Table 6.3.3 Development Network

Requirement	Specification	Purpose
Development Internet	25 Mbps minimum	Code repository access
Testing Connection	Stable broadband	App testing and debugging
Deployment Access	High-speed upload	App store publishing

6.4 Chapter Summary

The TriMinder application is designed to run efficiently on modern Android devices, requiring minimal hardware resources. The development environment uses industry-standard tools, including Flutter for cross-platform development, Supabase Pro tier for backend services, and SQLite for local data storage and offline functionality.

The dual-database approach ensures the app remains functional even with limited connectivity through SQLite local storage, while maintaining cloud synchronization capabilities through the Supabase Pro tier. Network requirements are optimized for both online and offline usage, ensuring robust performance across various connectivity scenarios.

The technical specifications support the project's goal of creating an accessible, gamified wellness application that can operate effectively across a wide range of Android devices while maintaining security, performance, and offline capability standards.

CHAPTER 7: BUDGET AND RESOURCES

7.1 Cost Breakdown

Summary of Direct vs Indirect Costs

- Direct costs are expenses explicitly tied to designing, developing, testing, and releasing TriMinder.
- Indirect costs support general academic activities (e.g., snacks) or provide contingency buffers.

Table 7.1.1 Itemized Budget Table

Classification	Item / Description	Unit Cost	Quantity	Sub-Total	Funding Source
DIRECT	Document printing (draft & final, A4)	₱5 /page	100 pages	₱500	Proponents
	Flash drive 32 GB (project backup)	₱270 /pc	1	₱270	Proponents
	Smart/TNT prepaid load cards (data for mobile testing)	₱100 /ea	10	₱1,000	Proponents
	Supabase “Pro” plan (1 month) for production DB	$25 ≈ ₱1,400	1 month	₱1,400	Proponents
	Contingency (10% of direct total)	—	—	₱317	Proponents
DIRECT SUB-TOTAL	—	—	—	₱3,487	—
INDIRECT	Meals & snacks for 30-participant usability test	₱100 per participant	30	₱3,000	Proponents

	Misc. office supplies (bond paper, ink, sticky notes)	₱500 lump-sum	1	₱500	Proponents
INDIRECT SUB-TOTAL	—	—	—	₱3,500	—
GRAND TOTAL	—	—	—	₱6,987	—

Exchange rate used: US$1 = ₱57 (July 2025).

7.2 Materials and Logistics

Table 7.2.1 Manpower and Roles

Role	**Primary Tasks**	**Assigned Member**	**Outside Expertise Needed**
Project Manager / Back-End Dev	Sprint planning, Supabase schema, API security	Jet Venson G. Romero	Supabase community forum
Mobile App Lead	Flutter UI, Android integration, unit testing	Russel Jhon C. Dasigan	None
UI/UX & QA	Wireframes, ISO/IEC 25010 usability testing, bug triage	Maria Angie C. Niez	Adviser feedback for heuristic evaluation
Capstone Adviser	Technical oversight, SDLC milestone approval	TBA	—
Volunteer Testers (30)	Usability & pilot study participants	Recruited students	—

Table 7.2.2 Equipment and Tools Inventory

Resource Type	Description	Availability	Remarks
Development laptops	Intel i5/8 GB RAM or higher (3 units)	Available	Personal devices of proponents
Android test phones	Android 10+, 4 GB RAM (3 units)	Available	Ensures wide OS coverage
High-speed internet	25 Mbps fiber at campus lab	Available	Primary coding & video-call channel
Mobile data packages	10 × Smart load ₱100	Procured	Real-world connectivity tests
Supabase backend	Pro tier	Provisioned	Scales for pilot users
GitHub private repo	Student discount (free)	Activated	Source control & CI
Design software	Figma (education plan)	Free	Interface prototypes & assets
Printing services	Local print shop offering ₱5/page	Accessible	Quick turnaround, Ormoc CBD
Binding service	Softbind ₱260/book average	Available	For final manuscript submission

Support Services and Facilities

- **Computer Laboratory (EVSU–Ormoc)**: Fallback build environment and group sprint reviews.
- **Campus Wi-Fi Hotspots**: Secondary internet channel for remote collaboration.
- **Local Print Shop**: Document printing and binding services for manuscripts and presentations.

Logistical Plan

- **Usability Testing**: Week 13 in campus lab (08:00–17:00); two 2-hour sessions with snacks.
- **Printing & Binding**: Week 17 at local print shop; prepare draft and final copies.
- **Final Deployment**: Week 15; Supabase Pro plan active and APK tested on target devices.

Resource Sufficiency Statement

Proponents' devices and free academic licenses cover all development needs. The allocated budget ensures at least one month of production hosting post-defense, allowing for final bug fixes and minor enhancements. Available equipment, workforce, and facilities confirm that TriMinder is both financially and logistically feasible within the academic term.

CHAPTER 8: CONCLUSION

This proposal addresses the critical problem of excessive and unregulated screen time among college students at Eastern Visayas State University (EVSU) Ormoc Campus, which negatively impacts their academic focus, sleep routines, physical activity, and mental wellness. The proposed system, titled "TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits," is designed to provide a reliable and user-friendly solution to improve digital discipline, wellness awareness, and peer-driven motivation through gamified engagement strategies.

The general objective of this project is to develop a comprehensive gamified screen time management and wellness monitoring application that offers interactive tracking, social engagement, and administrative oversight capabilities. Specifically, the system will provide core functionalities including real-time screen usage monitoring, gamification features such as XP points and digital discipline badges, friend-based ranking systems with peer interaction capabilities, wellness break notifications, and comprehensive administrative platforms for both SSG super admins and department-based sub-admins. These components are carefully aligned with the data gathered during the needs assessment phase, where over 200 EVSU Ormoc students expressed strong interest in gamified wellness solutions and peer-to-peer engagement features.

With the successful implementation of this project, the primary beneficiaries, such as EVSU Ormoc college students, are expected to gain improved digital self-discipline, enhanced awareness of screen usage patterns, reduced technology-related strain, and increased motivation through peer recognition and healthy competition. The gamified approach, incorporating XP points, badges, and daily rankings, will foster sustained engagement, while the friend system and profile interaction features will create a supportive community environment. Additionally, this system contributes to ongoing efforts toward digital transformation in higher education, aligning with institutional modernization goals and the Commission on Higher Education's advocacy for digital wellness and productivity in higher education institutions.

The feasibility of the project has been confirmed through careful planning, realistic cost estimation totaling ₱6,987, availability of appropriate development tools including Flutter SDK, Supabase Pro backend services, and SQLite for offline functionality, and the technical competency of the development team. The Agile Software Development Life Cycle methodology ensures iterative development with continuous user feedback integration across 18 weeks of structured development phases. All necessary resources, including development hardware, software licenses, network infrastructure, and stakeholder support mechanisms, have been identified and secured through a combination of personal resources and institutional facilities.

Following the approval of this proposal, the team will proceed to the comprehensive development phase, structured across seven distinct phases: Planning and Analysis, System Design, three iterative Development Sprints, System Integration and Testing, and finally Deployment and Documentation. This will include complete system architecture implementation, extensive usability testing based on ISO/IEC 25010 standards, user feedback integration through pilot deployments, and thorough quality assurance validation. The project team is committed to delivering a high-quality, gamified wellness solution that meets user needs, promotes sustainable digital habits, and supports the broader institutional goals of student wellness and academic excellence through innovative technology integration.

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APPENDICES

Appendix A – Survey Questionnaire

Survey Questionnaire: TriMinder User Insight Form

Title of the Study:

TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits among EVSU Ormoc Students

Purpose of the Study:

This survey aims to gather insights from EVSU Ormoc students regarding their screen time habits, preferences, and openness to using a gamified wellness tracking application. The data collected will help in designing *TriMinder*, a mobile application that promotes healthy digital habits through reminders, rewards, and peer interaction. Your honest responses will contribute to developing a system tailored to the needs and behaviors of students like you.

Instructions:

Please answer the following questions honestly. Your responses will remain confidential and will be used solely for research purposes.

Personal information

- Name: ___________________________
- Age: ____________________________
- Year Level: ______________________
- Section: _________________________

- Department: ____________________

Screen Time & Wellness Questions

1. On average, how many hours a day do you use your smartphone?

   ___ hours

1. What type of device do you primarily use?

   ☐ Android ☐ iOS (Apple) ☐ Other: ________________

2. Which apps do you use most frequently?

   List up to 3: _______________________________________

3. Do you currently track or monitor your screen time usage?

   ☐ Yes ☐ No

4. Have you ever felt that excessive screen time affects your sleep, productivity, or physical health?

   ☐ Yes ☐ No

5. Would you use an app that tracks your screen time and promotes wellness?

   ☐ Yes ☐ No

6. How important is it for you to receive notifications reminding you to take breaks (e.g., stretch, rest eyes)?

   ☐ 1 – Not important ☐ 2 ☐ 3 – Neutral ☐ 4 ☐ 5 – Very important

7. Would features like XP points, badges, or streak rewards motivate you to reduce screen time?

   ☐ Yes ☐ No

8. Do you prefer apps that allow personalization or goal setting (e.g., setting daily screen time limits)?

   ☐ Yes ☐ No

9. Would you be more motivated to manage your screen time if you were featured on a public ranking?

   ☐ Yes ☐ No

10. Would you like to see how your friends are performing in terms of digital discipline?

    ☐ Yes ☐ No

11. Would the ability to “like” or “unlike” friends’ performance profiles be meaningful for you?

    ☐ Yes ☐ No

12. Do you think friendly competition can help students be more mindful of their screen usage?

    ☐ Yes ☐ No

13. How likely are you to recommend a wellness-focused screen-tracking app to your classmates?

    ☐ 1 – Very unlikely ☐ 2 ☐ 3 – Neutral ☐ 4 ☐ 5 – Very likely

14. Any additional comments, feedback, or suggestions?

____

____

Consent Declaration

☐ I confirm that I have read the purpose of this survey and voluntarily agree to participate. I understand that all data will be treated confidentially and in an anonymous manner.

Signature (or initials): ______________________ **Date:** ________________

Appendix B – System Design Artifacts

B.1 Entity-Relationship Diagram (ERD)

This diagram represents the database structure for the TriMinder application, showing the relationships between users, screen time data, gamification elements, and guardian controls.

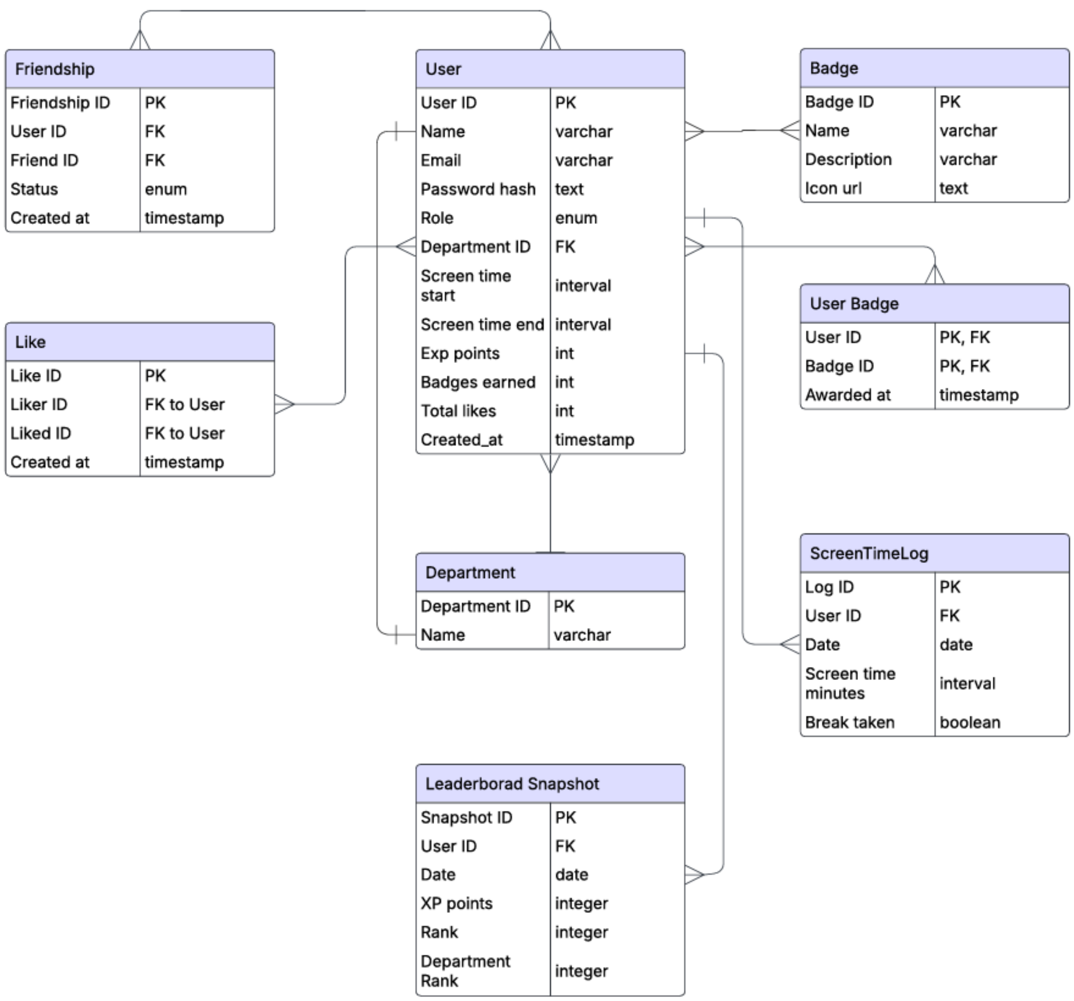

Figure B.1 *Entity-Relationship Diagram (ERD) for TriMinder*

Appendix C – Informed Consent Forms

C.1 Consent Form – Student Respondents

Informed Consent Form – Student Participants

Project Title:

TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits among EVSU Ormoc Students

Dear Participant,

You are invited to participate in a survey as part of our capstone research. The goal of this project is to develop a wellness and screen time monitoring app designed for students like you. The survey will take approximately 5–10 minutes.

Your participation is voluntary, and your identity will remain anonymous. Your answers will only be used for academic purposes and will help us improve the effectiveness of the app.

By participating, you agree to the following:

- I understand that I can withdraw at any time.
- I understand that all data will be kept confidential.
- I voluntarily agree to participate in the survey.

Signed: ________________________ **Date:** _________________

C.2 Consent Form – Student Organization Officer (Sub Admin)

Informed Consent Form – Department-Based Organization Officer

Dear Student Organization Officer,

We are conducting a capstone research project entitled *TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits Among EVSU Ormoc Students*. As a department-based officer, your insights will guide the sub-admin functionalities of our web system.

Your participation in this interview/survey is voluntary and will remain confidential. Your responses will help us understand how department organizations can promote digital wellness.

Please sign below to indicate your informed consent:

Name: ________________________ **Department:** ____________________

Signature: _____________________ **Date:** __________________________

C.3 Consent Form – Student Respondents

Informed Consent Form – Student Affairs and Services Office Representative

Dear SASO Representative,

You are invited to participate in our capstone research entitled TriMinder: Development of a Gamified Screen Time Management and Wellness Monitoring Application for Promoting Healthy Digital Habits Among EVSU Ormoc Students. Your role as a super admin is critical to designing the administrative features of the system.

We request your input through an interview or questionnaire. Your participation is voluntary, and all responses will be kept confidential.

Please indicate your consent by signing below:

Name: ________________________ **Department:** ____________________

Signature: _____________________ **Date:** __________________________

Appendix D – Gantt Chart

The TriMinder project follows a structured timeline encompassing seven key phases: Planning and Analysis (Weeks 1-2), System Design (Weeks 3-4), three sequential Development Sprints (Weeks 5-10), System Integration and Testing (Weeks 11-14), and Deployment and Documentation (Weeks 15-18). This phased approach ensures systematic progress from initial analysis through development and thorough testing to final deployment, optimizing project management and delivery within an 18-week timeframe.

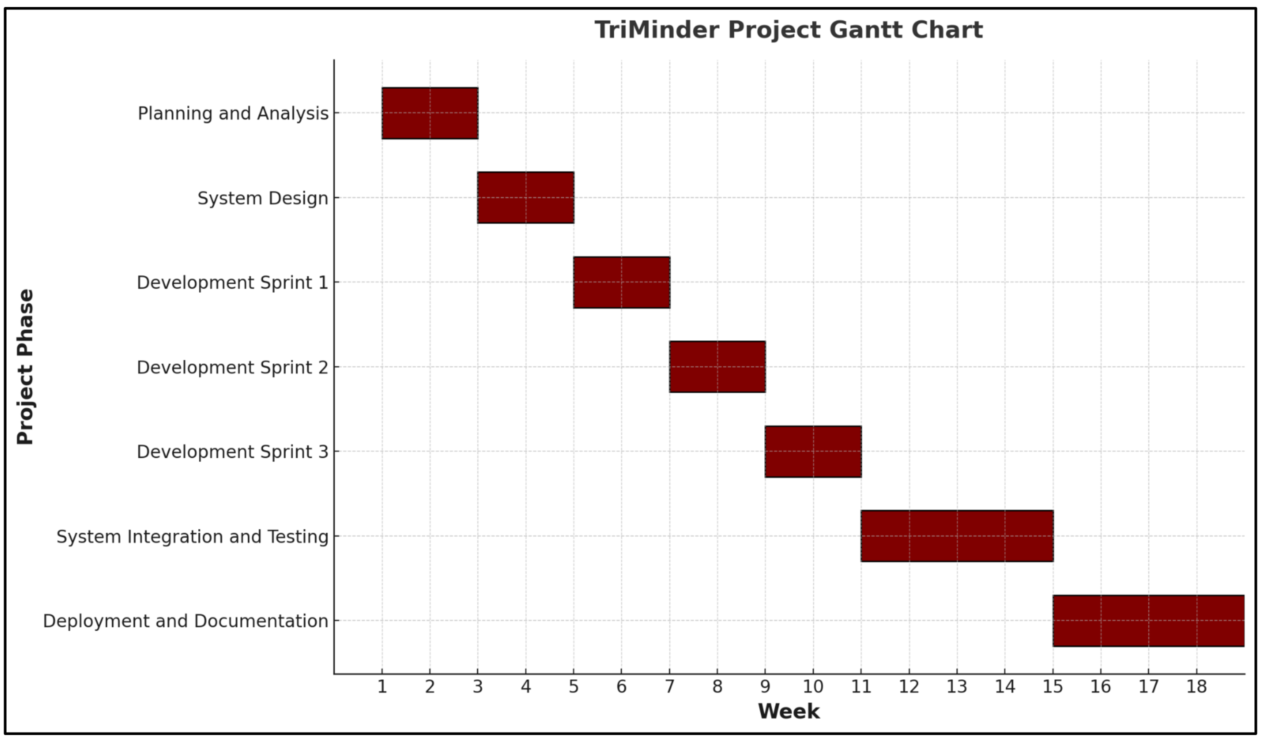

Figure D.1 *Gantt Chart for TriMinder Project Timeline*

Appendix E – Context-Level Data Flow Diagram (DFD) for TriMinder

This appendix presents the Context-Level Data Flow Diagram (DFD) for the TriMinder system. This facilitates user authentication, social interactions, and administrative functions, integrating features like screen time tracking, notifications, and gamification management. It manages data flow among users, departments, and system modules to provide a comprehensive notification and engagement platform.

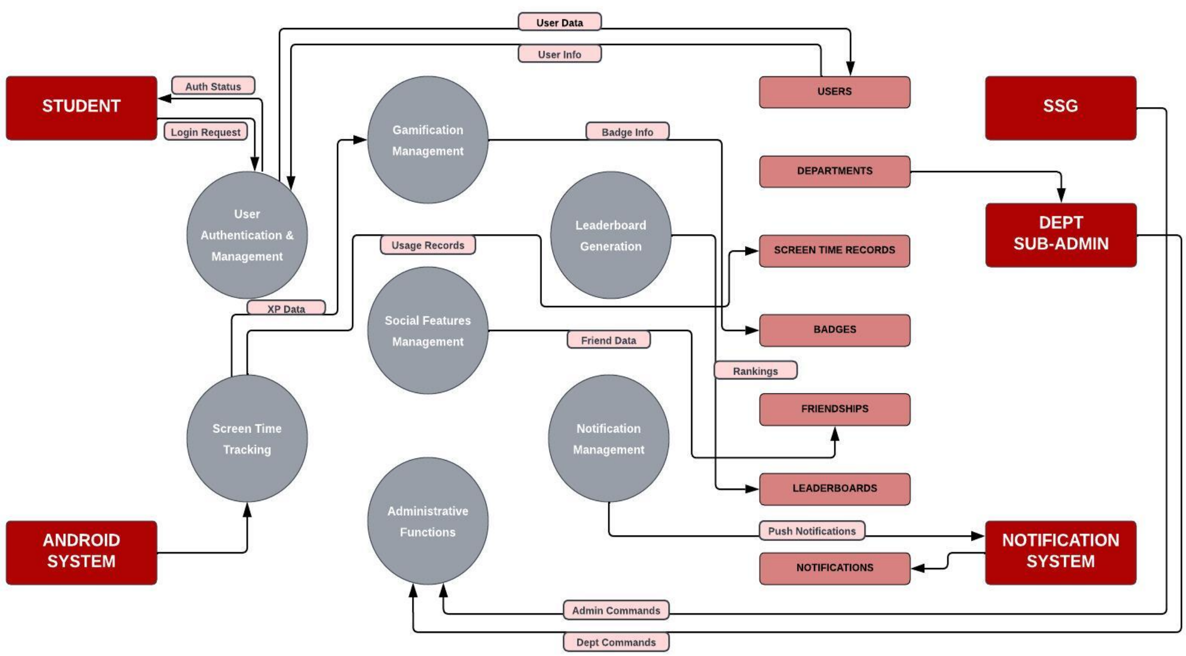

***Figure E.1** Context-Level Data Flow Diagram (DFD) for TriMinder*